

PAMANTASAN NG LUNGSOD NG PASIG
COLLEGE OF COMPUTER STUDIES
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

NEXUS PAYROLL: AN ERP MODULE

SYSTEM INTEGRATION AND ARCHITECTURE I
SUBJECT

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1.1 BACKGROUND

Nexus Payroll is a comprehensive, end-to-end payroll management system designed to automate and streamline the entire payroll process from employee time tracking to payslip generation. The system minimizes manual intervention by integrating advanced technologies such as facial recognition for time logging, automated attendance processing, and intelligent payroll computation.

The system serves as a centralized platform for Human Resources and Payroll departments, providing real-time visibility into employee attendance, leave management, payroll calculations, and comprehensive reporting. It ensures accuracy in salary computations while maintaining compliance with statutory requirements including SSS (Social Security System), PhilHealth, Pag-IBIG, and Withholding Tax contributions.

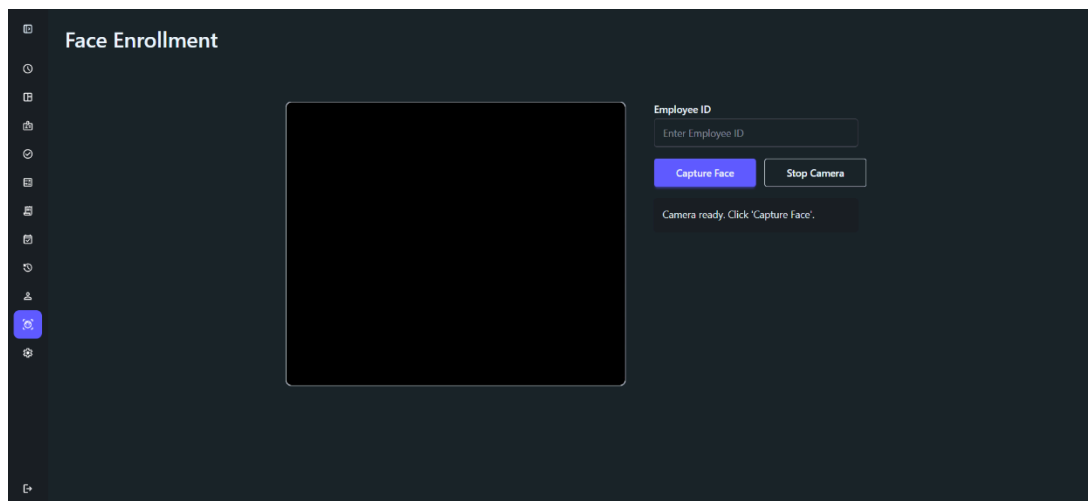
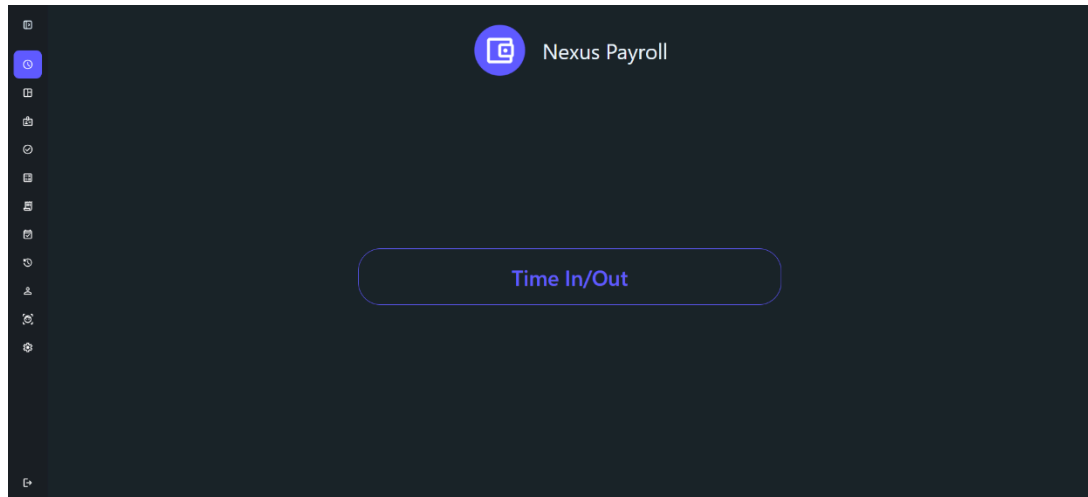
Built with a modular architecture, Nexus Payroll consists of core modules that work seamlessly together to provide a complete payroll solution. Each module is designed to handle specific aspects of the payroll workflow, from initial employee onboarding to final payslip distribution, ensuring data integrity and operational efficiency throughout the process.

1.2 SCOPE OF THE PROJECT

1.2.1 Time In/Out Management

- **Facial Recognition Time Logging:** Advanced face recognition technology using face-api.js library for secure and contactless employee identification
- **Facial recognition with real-time camera feed**
- **Real-time Face Detection:** WebGL-accelerated face detection and recognition using SSD MobileNet v1, Face Landmark 68, and Face Recognition models

- **Automatic Time Logging:** Intelligent system that automatically records time-in and time-out based on employee presence
- **Face Enrollment System:** Dedicated module for enrolling new employees' facial data into the system
- **Confirmation Overlays:** Visual confirmation displays showing employee name, ID, action (TIME IN/OUT), and timestamp



1.2.2 Employee Information & Records Management

- **Comprehensive Employee Database:** Centralized repository for all employee information
- **Personal Information Management:**
 - Employee ID, full name, contact details, address
 - Profile management and updates
- **Employment Details Tracking:**
 - Position and department assignments
 - Hire date and employment status
 - Official time schedules
- **Data Grid Interface:** Advanced AG Grid implementation for efficient data viewing, searching, and filtering
- **CRUA Operations:** Full Create, Read, Update, Archive functionality for employee records

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1.2.3 Attendance, Leave, and Absence Management

- **Raw Time Logs Processing:** Conversion of raw time entries into verified timesheets
- **Attendance Summary Generation:** Automated computation of attendance metrics per cutoff period
- **Hours Calculation:**
 - Regular hours computation
 - Overtime hours calculation
 - Undertime detection and tracking
- **Attendance Anomaly Detection:**
 - Late arrival identification
 - Absence tracking
 - Undertime monitoring
- **Leave Management:**
 - Paid and unpaid leave processing
 - Leave application and approval workflow
 - Automatic payroll integration for leave deductions
- **DTR (Daily Time Record) Generation:** Automated generation of Daily Time Records in PDF format
- **Cutoff Period Management:** Flexible cutoff period configuration and filtering
- **Data Export:** CSV export functionality for attendance records
- **Dual View System:** Toggle between Raw Time Logs and Attendance Summary views

Attendance Management

Raw Time Logs

Attendance Summary

Q Search

☐	Date	Employee ID	Last Name	First Name	Middle Name	Official Time	Time In
☐	2025-10-20	5	Ray	Alistair	E	07:00:00 - 16:00:00	2025-10-20 07:01:00
☐	2025-10-20	1	Vance	Elara	C	09:00:00 - 18:00:00	2025-10-20 08:58:00
☐	2025-10-20	2	Kall	Jaxon	B	09:00:00 - 18:00:00	2025-10-20 09:00:00
☐	2025-10-20	3	Chen	Lian	A	10:00:00 - 19:00:00	2025-10-20 09:50:00
☐	2025-10-20	4	Singh	Maya	D	10:00:00 - 19:00:00	2025-10-20 10:20:00
☐	2025-10-21	5	Ray	Alistair	E	07:00:00 - 16:00:00	2025-10-21 07:05:00
☐	2025-10-21	2	Kall	Jaxon	B	09:00:00 - 18:00:00	2025-10-21 08:55:00
☐	2025-10-21	1	Vance	Elara	C	09:00:00 - 18:00:00	2025-10-21 09:15:00
☐	2025-10-21	4	Singh	Maya	D	10:00:00 - 19:00:00	2025-10-21 09:55:00
☐	2025-10-21	3	Chen	Lian	A	10:00:00 - 19:00:00	2025-10-21 10:02:00
☐	2025-11-01	11	Tester	Trigger	Z	09:00:00 - 18:00:00	2025-11-01 09:00:00

Generate DTR

Generate CSV

Leave Management

Q Search

Delete

Add New

☐	Employee ID	Name	Position	Department	Leave Duration	Type	Status	Leave Balance
☐	3	Lian A Chen	Staff Accountant	Accounting	2 day(s)	Personal Leave	Paid	View More
☐	1	Elara C Vance	Senior Developer	Information Technology	6 day(s)	Vacation Leave	Unpaid	View More
☐	11	Trigger Z Tester	Senior Developer	Information Technology	4 day(s)	Personal Leave	Paid	View More
☐	2	Jaxon B Kall	HR Specialist	Human Resources	4 day(s)	Sick Leave	Unpaid	View More
☐	2	Jaxon B Kall	HR Specialist	Human Resources	2 day(s)	Sick Leave	Unpaid	View More
☐	7	Oscar G Perez	Staff Accountant	Accounting	2 day(s)	Emergency Leave	Unpaid	View More
☐	4	Maya D Singh	Sales Manager	Sales	5 day(s)	Vacation Leave	Paid	View More
☐	12	Christophers Angga Quinto	Senior Developer	Information Technology	3 day(s)	Sick Leave	Unpaid	View More

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Generate CSV

1.2.4 Payroll Computation & Compliance

- **Automated Salary Computation:**
 - Base salary calculation based on employee rates
 - Gross pay computation
 - Net pay calculation
- **Deduction Management:**
 - Lateness deductions

- o Undertime deductions
- o Absence deductions

Statutory Compliance:

- o SSS contribution calculations using updated contribution tables
- o PhilHealth premium computations
- o Pag-IBIG contribution deductions
- o Withholding Tax (BIR) calculations
- **Payroll Register Generation:** Comprehensive payroll register with gross pay, all deductions, and net pay
- **Cutoff Period Integration:** Payroll computation aligned with defined cutoff periods
- **Data Grid Interface:** Advanced grid system for payroll data management and review

The screenshot displays the 'Payroll Computation' interface. At the top, there is a search bar and a 'Cutoff Period' dropdown menu set to 'All Cutoff'. Below this is a table with the following columns: Employee ID, Name, Gross, Total Deductions, Cutoff Period, Department, and Actions. The table contains 10 rows of employee data. Each row has a 'View More' button in the Actions column. At the bottom right, there is a 'Generate CSV' button.

Employee ID	Name	Gross	Total Deductions	Cutoff Period	Department	Actions
1000123	Juan Dela Cruz	38000	3200	Oct 1 - 15, 2025	Finance	View More
1000456	Maria Santos	42000	3800	Oct 16 - 31, 2025	HR	View More
1000789	Jose Ramirez	39500	2900	Nov 1 - 15, 2025	IT	View More
1001024	Ana Bautista	41000	3500	Oct 1 - 15, 2025	Sales	View More
1001345	Pedro Cruz	36500	2700	Oct 16 - 31, 2025	Operations	View More
1001567	Rosa Flores	39000	3100	Nov 1 - 15, 2025	Finance	View More
1001890	Carlos Navarro	45500	4100	Oct 1 - 15, 2025	IT	View More
1002234	Elena Mendoza	37000	2950	Oct 16 - 31, 2025	HR	View More
1002456	Ramon Reyes	48000	4300	Nov 1 - 15, 2025	Sales	View More
1002789	Lucia Morales	40500	3100	Oct 1 - 15, 2025	Operations	View More

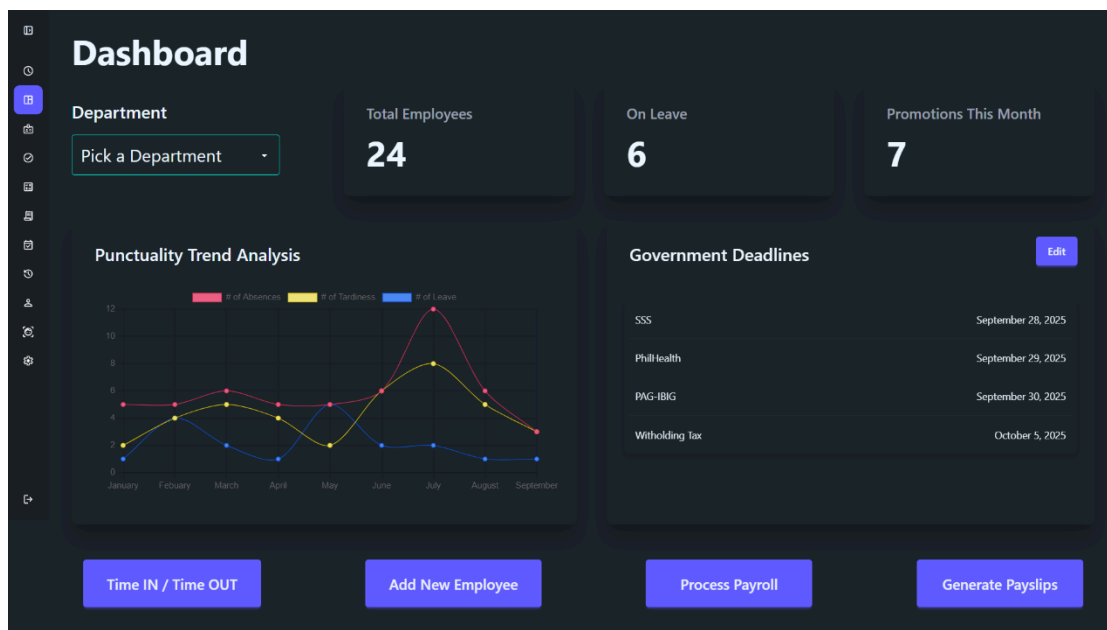
1.2.5 Payslip Generation & Salary Disbursement

- **Digital Payslip Generation:** Automated generation of professional payslips
- **PDF Export:** High-quality PDF generation using jsPDF library
- **Batch Processing:** Generate multiple payslips simultaneously (2 payslips per page)

- **Detailed Breakdown Display:**
 - Basic pay information
 - Overtime earnings
 - All deductions (SSS, PhilHealth, Pag-IBIG, Withholding Tax, loans, etc.)
 - Net pay calculation
- **Attendance Summary Integration:** Payslips include attendance and leave summaries for reference
- **Printable Format:** Professional layout suitable for printing and digital distribution
- **Employee Selection:** Flexible selection of employees for payslip generation

1.2.6 Reports, Analytics, and Audit Trail

- **Dashboard Analytics:**
 - Real-time employee statistics
 - Department-wise filtering
 - Quick statistics display (Total Employees, On Leave, Promotions)
- **Punctuality Trend Analysis:** Visual charts showing attendance trends over time
- **Data Visualization:** Chart.js integration for interactive charts and graphs
- **Comprehensive Reporting:** Various report types for management and finance use
- **Audit Trail:** Secure logging of all system activities and changes



Audit Trail

Filter by actions: Generate CSV Search

Name	Action	Description	Module Affected	Record ID	User Agent	Time
Christopher Quinto	View	Exported attendance dat...	Attendance Management	88	Mozilla/5.0 (Windows NT...	2025-11-30 12:30:59
Christopher Quinto	View	Exported attendance dat...	Attendance Management	87	Mozilla/5.0 (Windows NT...	2025-11-30 12:29:46
Christopher Quinto	Create	Created Sick Leave reque...	Leave Management	86	Mozilla/5.0 (Windows NT...	2025-11-29 03:14:54
Christopher Quinto	Create	Created Emergency Leav...	Leave Management	85	Mozilla/5.0 (Windows NT...	2025-11-29 03:14:20
Christopher Quinto	Create	Created Emergency Leav...	Leave Management	84	Mozilla/5.0 (Windows NT...	2025-11-29 03:14:18
Ipei Goto	View	Exported leave balance d...	Leave Management	82	Mozilla/5.0 (Windows NT...	2025-11-28 18:40:23
Ipei Goto	View	Generated leave balance ...	Leave Management	81	Mozilla/5.0 (Windows NT...	2025-11-28 18:40:18
Ipei Goto	Create	Created Personal Leave r...	Leave Management	80	Mozilla/5.0 (Windows NT...	2025-11-28 18:39:34
Ipei Goto	Edit	Updated employee infor...	Employee Management	79	Mozilla/5.0 (Windows NT...	2025-11-28 18:28:38
Ipei Goto	Edit	Updated employee infor...	Employee Management	78	Mozilla/5.0 (Windows NT...	2025-11-28 18:28:23
Ipei Goto	Create	Added new employee: M...	Employee Management	77	Mozilla/5.0 (Windows NT...	2025-11-28 18:26:11
Ipei Goto	View	Exported attendance dat...	Attendance Management	76	Mozilla/5.0 (Windows NT...	2025-11-28 18:22:30
Ipei Goto	View	Generated DTR PDF for S...	Attendance Management	75	Mozilla/5.0 (Windows NT...	2025-11-28 18:17:00

1.2.7 User Management & Authentication

- **User Authentication:** Secure login system using Supabase Auth
- **User Account Management:**
 - Add new users
 - User role management
 - Password reset functionality
- **Session Management:** Secure session handling and authentication state
- **Access Control:** Role-based access control system

User Management

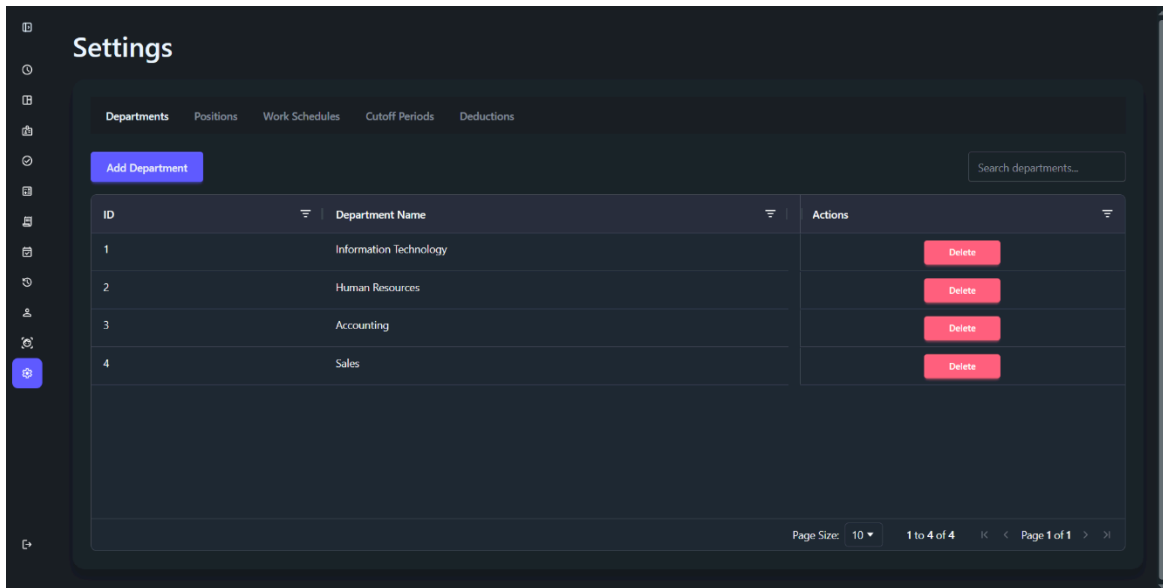
Search Add New

Status	Actions	Username	First Name	Last Name	User Type	Employee ID	Email	Created At	Last Sign In
Active	Reset	Elara	Elaris	Vance	Employee	1	vance_elara@gmail.com	11/22/2025, 9:48:29 AM	11/28/2025, 9:56:23 PM
Active	Reset	Juan	Juan	Dela Cruz	Payroll Staff	-	delacruz_juan@gmail.c...	11/22/2025, 9:46:28 AM	11/22/2025, 9:49:14 AM
Active	Reset	keith_tornea	Keith	Andrea Tornea	Admin	-	tornea.keithandrea@g...	11/18/2025, 6:27:10 PM	11/25/2025, 5:19:30 PM
Active	Reset	sofia_manuel	Sofia	Manuel	Admin	-	manuelsofia@gmail.com	11/16/2025, 11:05:53 ...	11/16/2025, 11:06:28 ...
Active	Reset	Ipei	Ipei	Goto	Admin	-	ipeigotou@gmail.com	11/13/2025, 12:26:21 ...	11/13/2025, 12:27:53 ...

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1.2.8 System Settings & Configuration

- **Department Management:** CRUD operations for departments
- **Position Management:** Job position definitions and management
- **Official Time Schedule Configuration:** Work schedule setup and management
- **Cutoff Period Settings:** Payroll cutoff period configuration



1.3 TECHNOLOGIES USED

Core Technologies

- **HTML5:** Semantic markup for all application pages
- **JavaScript (ES6+):** Modern JavaScript with ES6 modules for application logic
- **CSS3:** Advanced styling and animations

CSS Frameworks & Libraries

- **TailwindCSS v4.1.14:** Utility-first CSS framework for rapid UI development
 - o PostCSS integration for CSS processing
 - o Custom configuration via `postcss.config.cjs`
 - o Responsive design utilities

- **DaisyUI v5.3.0:** Component library built on TailwindCSS
 - Pre-built UI components (buttons, modals, forms, cards, etc.)
 - Theme system support
 - Accessibility features

JavaScript Libraries & Frameworks

- **AG Grid Community v34.2.0:** Open-source data grid for displaying tabular data
 - Advanced filtering and sorting
 - Column management
 - Data export capabilities
- **AG Grid Enterprise v34.2.0:** Enterprise features for data grids
 - Enhanced functionality
 - Advanced data manipulation
- **Chart.js:** JavaScript charting library for data visualization
 - Line charts for trend analysis
 - Interactive charts
 - Responsive design
- **face-api.js (@vladmandic/face-api v1.5.7):** Facial recognition library
 - Face detection using SSD MobileNet v1
 - Face landmark detection (68 points)
 - Face recognition and matching
 - WebGL acceleration for performance
- **jsPDF v2.5.1:** PDF generation library
 - Client-side PDF creation
 - Multi-page document support
 - Custom formatting and layouts

Build Tools

- **PostCSS v8.5.6**: CSS processing tool
- **PostCSS CLI v11.0.1**: Command-line interface for PostCSS
- **Autoprefixer v10.4.21**: Automatic vendor prefixing for CSS
- **@tailwindcss/postcss v4.1.14**: TailwindCSS PostCSS plugin

Development Tools

- **Node.js & npm**: Package management and build scripts
- **Git & GitHub**: Version control and collaboration

Back End

Backend-as-a-Service (BaaS)

- **Supabase**: Complete backend solution providing:
 - PostgreSQL database
 - Real-time subscriptions
 - Authentication service
 - RESTful API
 - Row Level Security (RLS)
 - Storage capabilities

Supabase Services Used

- **Supabase Auth**:
 - User authentication (email/password)
 - Session management
 - User registration
 - Password reset functionality
- **Supabase Database (PostgreSQL)**:
 - Relational database for all application data
 - Custom SQL functions and stored procedures

- o Database views for optimized queries
- o Foreign key relationships
- **Supabase Client Library (@supabase/supabase-js v2):**
 - o JavaScript client for Supabase API
 - o Real-time data subscriptions
 - o Query builder
 - o File upload/download

External APIs & CDNs

1. Supabase API:

- o **Authentication:** JWT-based authentication
- o **Endpoints:**
 - REST API for database operations
 - Auth API for user management
 - Storage API for file operations
- o **Real-time API:** WebSocket connections for live updates

2. Face-API.js CDN:

- o **URL:** <https://cdn.jsdelivr.net/npm/@vladmandic/face-api@1.5.7/>
- o **Purpose:** Facial recognition models and library
- o **Models Used:**
 - SSD MobileNet v1 (face detection)
 - Face Landmark 68 Net (facial landmarks)
 - Face Recognition Net (face descriptors)

3. Chart.js CDN:

- o **URL:** <https://cdn.jsdelivr.net/npm/chart.js>
- o **Purpose:** Data visualization and charting

4. **jsPDF CDN:**

- o **URL:** <https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/jspdf.umd.min.js>
- o **Purpose:** Client-side PDF generation

5. **Supabase JavaScript Client CDN:**

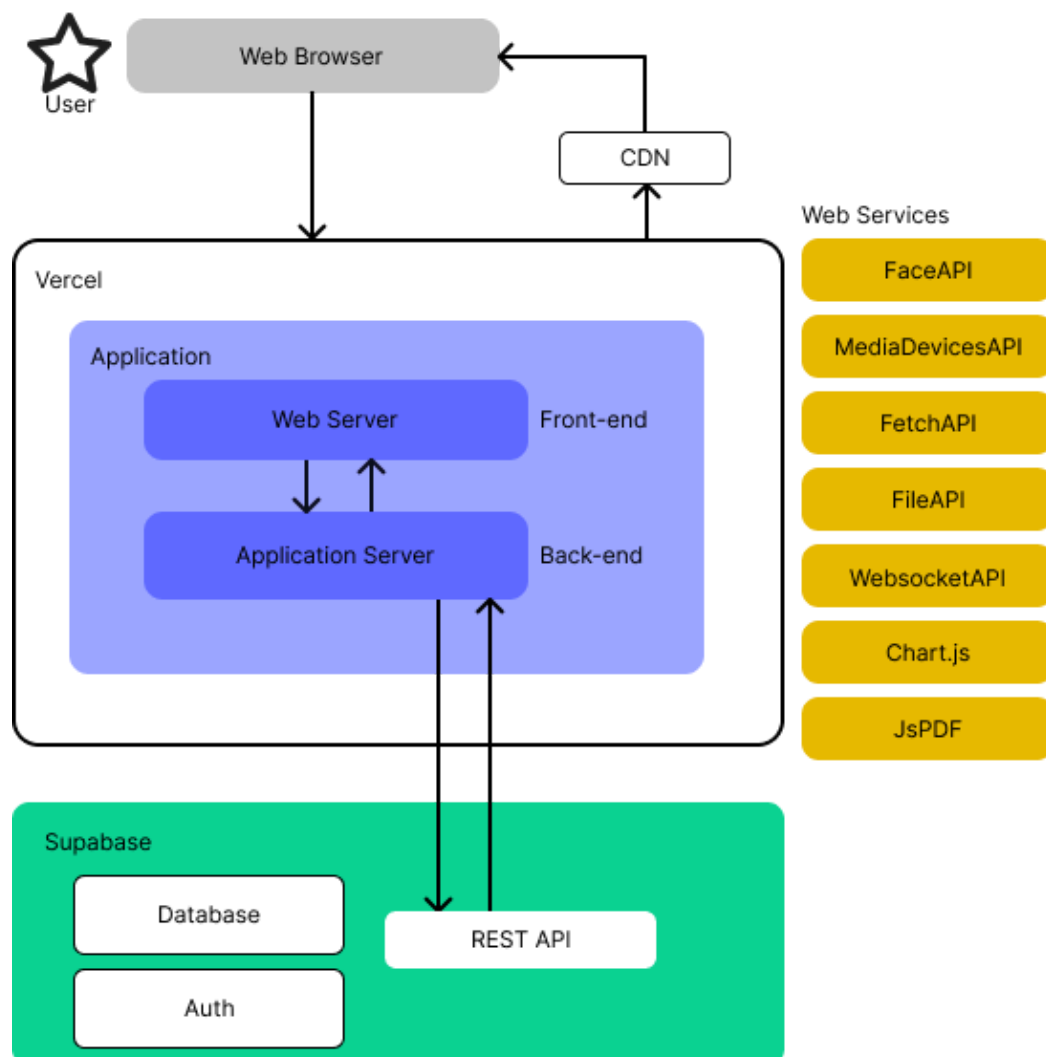
- o **URL:** <https://cdn.jsdelivr.net/npm/@supabase/supabase-js@2>
- o **Purpose:** Supabase client library

Browser APIs

- **MediaDevices API:** Camera access for facial recognition
- **WebGL API:** GPU acceleration for face detection
- **File API:** File handling for exports
- **Fetch API:** HTTP requests to Supabase
- **WebSocket API:** Real-time data subscriptions (via Supabase)

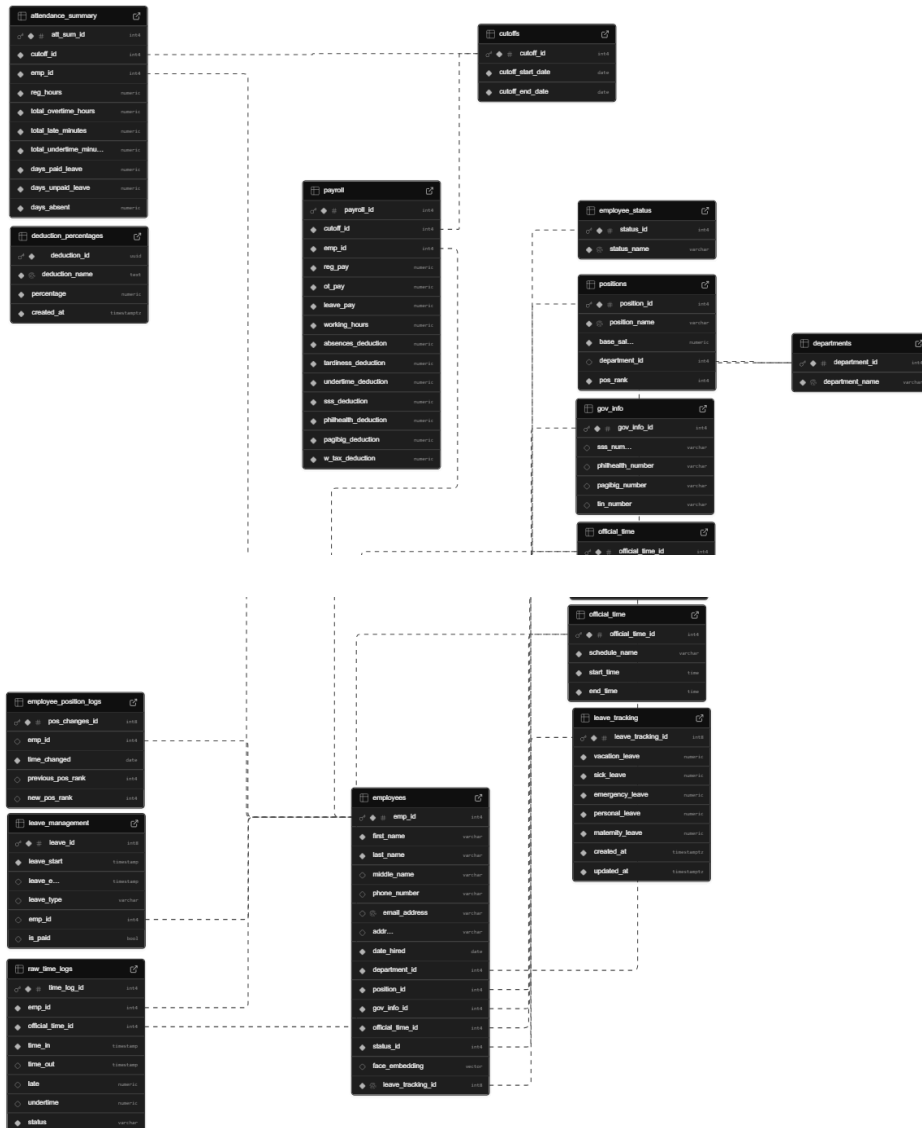
1.4 ARCHITECTURAL DESIGN

The web application is deployed on Vercel with Supabase as the backend infrastructure. When a user accesses the application through their web browser, requests are routed through a CDN for optimized content delivery to the Vercel-hosted application layer. Within Vercel, the application is divided into a front-end Web Server and a back-end Application Server that communicate bidirectionally to handle user interactions and business logic. The Application Server connects to Supabase, which provides both database storage and authentication services, as well as a REST API layer for data operations. The application integrates with multiple external web services including FaceAPI and MediaDevicesAPI for media handling, FetchAPI and FileAPI for data and file operations, WebsocketAPI for real-time communication, and Chart.js and JSPDF for data visualization and document generation.



1.5 DATABASE SCHEMA

This database schema supports a comprehensive payroll system. It consists of multiple interconnected tables that manage employee information, attendance, time logs, payroll computation, leave tracking, and audit records.



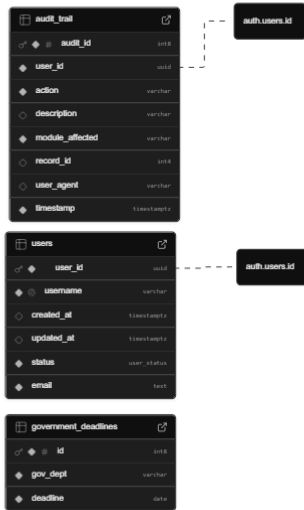


Table: employees

Central table storing employee information and relationships to other core entities.

Column	Type	Constraints	Description
emp_id	int	Primary Key, Identity	Unique employee identifier
first_name	varchar	NOT NULL	Employee first name
last_name	varchar	NOT NULL	Employee last name
middle_name	varchar	NULL	Employee middle name
phone_number	varchar	NULL	Contact phone number
email_address	varchar	UNIQUE	Employee email
address	varchar	NULL	Physical address

date_hired	date	NOT NULL	Employment start date
department_id	int	NOT NULL, FK	Reference to departments
position_id	int	NOT NULL, FK	Reference to positions
gov_info_id	int	NOT NULL, FK	Reference to gov_info
official_time_id	int	NOT NULL, FK	Reference to official_time
status_id	int	NOT NULL, DEFAULT 1, FK	Reference to employee_status
face_embedding	user-defined	NULL	Biometric data for facial recognition
leave_tracking_id	bigint	NOT NULL, UNIQUE, FK	Reference to leave_tracking

Table: departments

Stores organizational departments.

Column	Type	Constraints	Description
department_id	int	PRIMARY KEY, IDENTITY	Unique department identifier
department_name	varchar	NOT NULL, UNIQUE	Department name

Table: positions

Defines job positions with salary information and hierarchical ranking.

Column	Type	Constraints	Description
position_id	int	PRIMAR KEY, IDENTITY	Unique position identifier

position_name	varchar	NOT NULL, UNIQUE	Position title
base_salary	numeric	NOT NULL	Base salary for position
department_id	int	FK	Reference to departments
pos_rank	int	NOT NULL, DEFAULT 1	Hierarchical rank

Table: employee_status

Lookup table for employee status values.

Column	Type	Constraints	Description
status_id	int	PRIMARY KEY, IDENTITY	Unique status identifier
status_name	varchar	NOT NULL, UNIQUE	Status name(e.g., Active, Inactive)

Table: official_time

Defines work schedules with start and end times.

Column	Type	Constraints	Description
official_time_id	int	PRIMARY KEY, IDENTITY	Unique schedule identifier
schedule_name	varchar	NOT NULL	Schedule name
start_time	time	NOT NULL	Work start time
end_time	time	NOT NULL	Work end time

Table: raw_time_logs

Records individual employee clock-in/clock-out events.

Column	Type	Constraints	Description
time_log_id	int	PRIMARY KEY, IDENTITY	Unique log identifier
emp_id	int	NOT NULL, FK	Reference to employees
official_time_id	int	NOT NULL, FK	Reference to official_time
time_in	timestamp	NOT NULL	Clock-in timestamp
time_out	timestamp	NULL	Clock-out timestamp
late	numeric	NULL	Late minutes
undertime	numeric	NULL	Undertime minutes
status	varchar	NULL	Log status

Table: attendance_summary

Aggregated attendance data per employee per cutoff period.

Column	Type	Constraints	Description
att_sum_id	int	PRIMARY KEY, IDENTITY	Unique summary identifier
cutoff_id	int	NOT NULL, FK	Reference to cutoffs
emp_id	int	NOT NULL, FK	Reference to employees
reg_hours	numeric	NOT NULL, DEFAULT 0.00	Regular hours worked
total_overtime_hours	numeric	NOT NULL, DEFAULT 0.00	Total overtime hours
total_late_minutes	numeric	NOT NULL, DEFAULT 0.00	Total late minutes
total_undertime_minutes	numeric	NOT NULL, DEFAULT 0.00	Total undertime minutes

days_paid_leave	numeric	NOT NULL, DEFAULT 0.00	Paid leave days taken
days_unpaid_leave	numeric	NOT NULL, DEFAULT 0.00	Unpaid leave days taken
days_absent	numeric	NOT NULL, DEFAULT 0.00	Absent days

Table: leave_tracking

Records individual leave requests and usage.

Column	Type	Constraints	Description
leave_tracking_id	bigint	PRIMARY KEY, IDENTITY	Unique tracking identifier
vacation_leave	numeric	NOT NULL, DEFAULT 15.0	Available vacation leave days
sick_leave	numeric	NOT NULL, DEFAULT 15.0	Available sick leave days
emergency_leave	numeric	NOT NULL, DEFAULT 5.0	Available emergency leave days
personal_leave	numeric	NOT NULL, DEFAULT 3.0	Available personal leave days
maternity_leave	numeric	NOT NULL, DEFAULT 105.0	Available maternity leave days
created_at	timestamps	NOT NULL, DEFAULT now()	Record creation timestamp
updated_at	timestamps	NOT NULL, DEFAULT now()	Record update timestamp

Table: leave_management

Records individual leave requests and usage.

Column	Type	Constraints	Description
leave_id	bigint	PRIMARY KEY, IDENTITY	Unique leave identifier
leave_start	timestamp	NOT NULL	Leave start date/time
leave_end	timestamp	NULL	Leave end date/time
leave_type	varchar	NULL	Type of leave taken
emp_id	int	FK	Reference to employees
is_paid	boolean	NULL	Whether leave is paid

Table: cutoffs

Defines payroll periods.

Column	Type	Constraints	Description
cutoff_id	integer	PRIMARY KEY, IDENTITY	Unique cutoff identifier
cutoff_start_date	date	NOT NULL	Period start date
cutoff_end_date	date	NOT NULL	Period end date

Table: payroll

Stores calculated payroll data for each employee per cutoff period.

Column	Type	Constraints	Description
payroll_id	int	PRIMARY KEY, IDENTITY	Unique payroll identifier

cutoff_id	int	NOT NULL, FK	Reference to cutoffs
emp_id	int	NOT NULL, FK	Reference to employees
reg_pay	numeric	NOT NULL	Regular pay amount
ot_pay	numeric	NOT NULL	Overtime pay amount
leave_pay	numeric	NOT NULL	Leave pay amount
working_hours	numeric	NOT NULL	Total working hours
absences_deduction	numeric	NOT NULL	Deduction for absences
tardiness_deduction	numeric	NOT NULL	Deduction for tardiness
undertime_deduction	numeric	NOT NULL	Deduction for undertime
sss_deduction	numeric	NOT NULL	SSS contribution deduction
philhealth_deduction	numeric	NOT NULL	PhilHealth contribution deduction
pagibig_deduction	numeric	NOT NULL	Pag-IBIG contribution deduction
w_tax_deduction	numeric	NOT NULL	Withholding tax deduction

Table: deduction_percentage

Stores configurable deduction rates for government contributions.

Column	Type	Constraints	Description
deduction_id	uuid	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique deduction identifier
deduction_name	text	NOT NULL, UNIQUE	Deduction name

percentage	numeric	NOT NULL, CHECK (0-1)	Deduction percentage (0-1)
created_at	timestamps	NOT NULL, DEFAULT now()	Record creation timestamp

Table: gov_info

Stores government-issued identification numbers.

Column	Type	Constraints	Description
gov_info_id	int	PRIMARY KEY, IDENTITY	Unique identifier
sss_number	varchar	NULL	Social Security System number
philhealth_number	varchar	NULL	PhilHealth identification number
pagibig_number	varchar	NULL	Pag-IBIG identification number
tin_number	varchar	NULL	Tax Identification Number

Table: government_deadlines

Tracks government reporting and payment deadlines.

Column	Type	Constraints	Description
id	bigint	PRIMARY KEY, IDENTITY	Unique deadline identifier
gov_dept	varchar	NOT NULL	Government department name
deadline	date	NOT NULL	Deadline date

Table: audit_trail

Logs all system actions for security and compliance.

Column	Type	Constraints	Description
audit_id	bigint	PRIMARY KEY, IDENTITY	Unique audit identifier
user_id	uuid	NOT NULL, FK	Reference to auth.users
action	varchar	NOT NULL	Action performed
description	varchar	NULL	Action description
module_affected	varchar	NOT NULL	System module affected
record_id	int	NULL	ID of affected record
user_agent	varchar	NULL	User's browser/client info
timestamp	timestamps	NOT NULL, DEFAULT now()	Action timestamp

Table: employee_position_logs

Tracks historical position changes for employees.

Column	Type	Constraints	Description
pos_changes_id	bigint	PRIMARY KEY, IDENTITY	Unique log identifier
emp_id	int	FK	Reference to employees
time_changed	date	NOT NULL	Date of position change
previous_pos_rank	int	NULL	Previous position rank
new_pos_rank	int	NULL	New position rank

Table: users

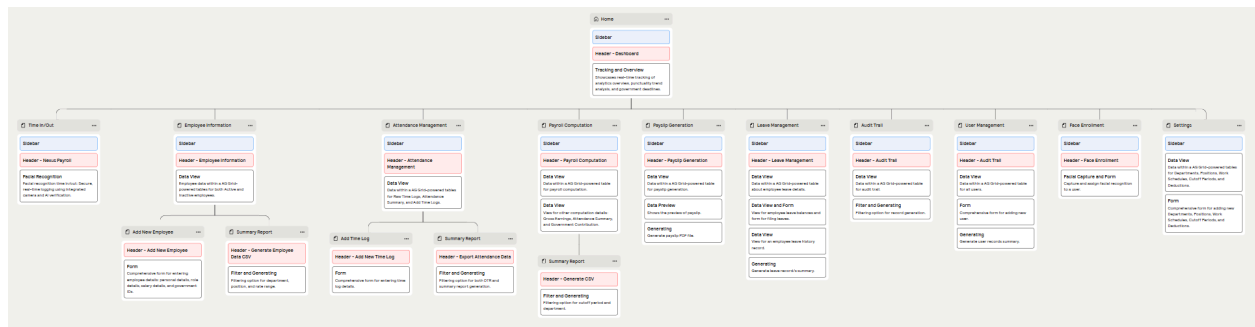
Application user accounts linked to authentication system.

Column	Type	Constraints	Description
user_id	uuid	PRIMARY KEY, FK	Reference to auth.users
username	varchar	NOT NULL, UNIQUE	Username
created_at	timestamps	DEFAULT now()	Account creation timestamp
updated_at	timestamps	DEFAULT now()	Account update timestamp
status	USER-DEFINED	NOT NULL, DEFAULT 'active'	User account status
email	text	NOT NULL	User email address

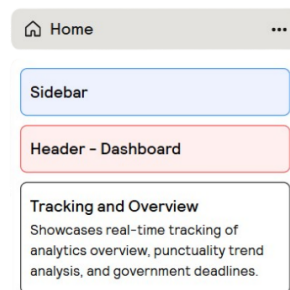
1.6 SITEMAP

This sitemap outlines the structure of the Nexus Payroll System, showing its main modules which are Dashboard, Payroll, Payslip Generation, Attendance, Employee Management, Leave, Audit Trail, User Management, Face Enrollment, and Settings. Each module includes its key functions such as data views, forms, filters, and record or PDF generation, providing a clear view of user flow and system organization.

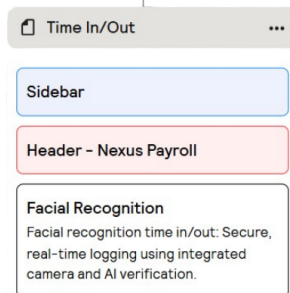
1.6.1 NEXUS-Payroll Sitemap



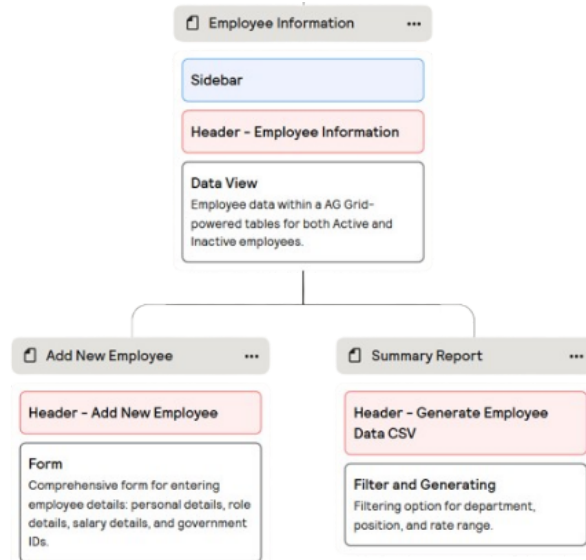
1.6.2 Home/Dashboard



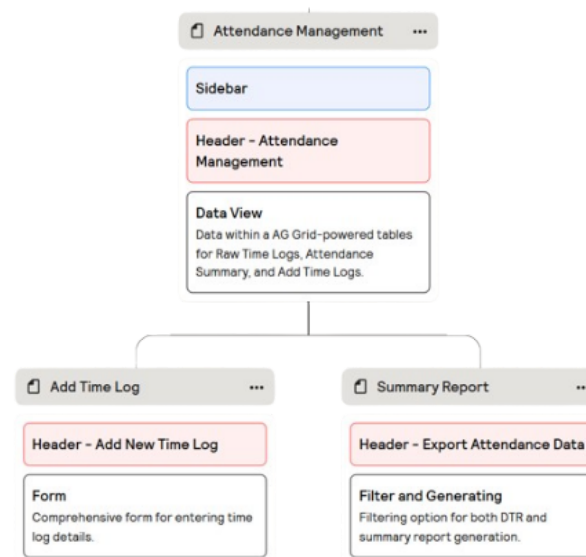
1.6.3 Time In/Out



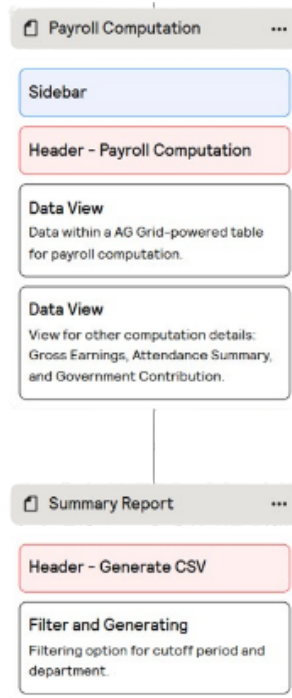
1.6.4 Employee Information



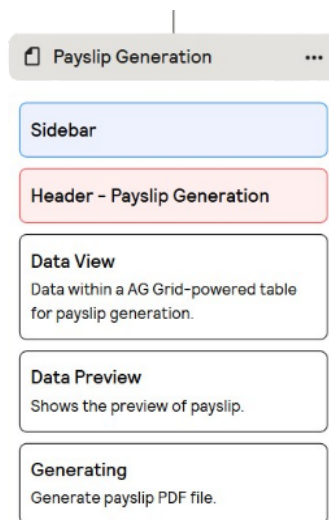
1.6.5 Attendance Management



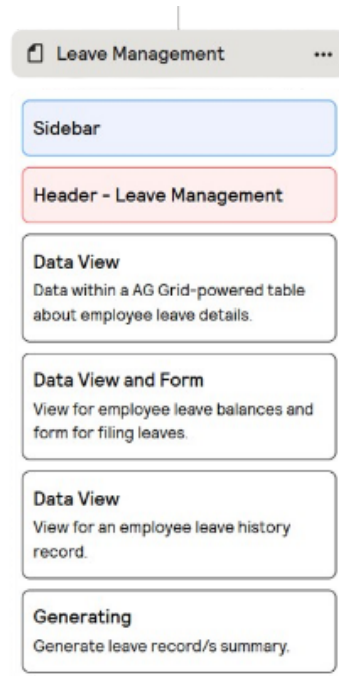
1.6.6 Payroll Computation



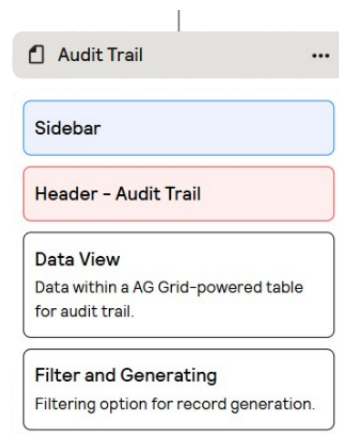
1.6.7 Payslip Generation



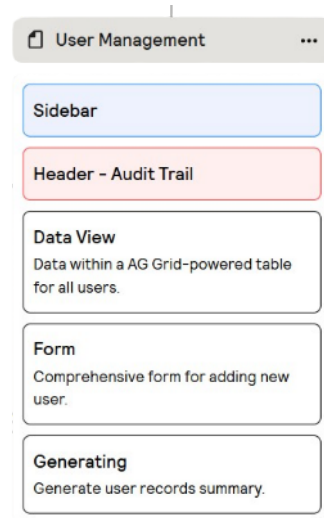
1.6.8 Leave Management



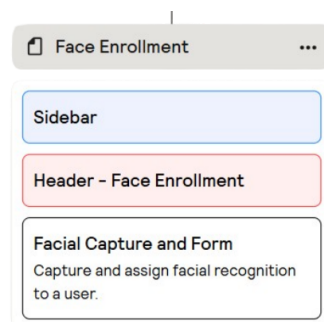
1.6.9 Audit Trail



1.6.10 User Management



1.6.11 Face Enrollment



1.6.12 Settings

